NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

METRIC

MIL-PRF-12039D(OS)
14 November 1990
SUPERSEDING
MIL-L-12039C(OS)
8 May 1974

PERFORMANCE SPECIFICATION

LITHIUM HYDRIDE, TECHNICAL (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes one grade of powdered lithium hydride.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandind Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6810

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FED-STD-313 Material Safety Data Sheets, Preparation and Submission of

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation (see 6.2).

CODE OF FEDERAL REGULATIONS

29 CFR Parts 1900-1910 Labor
49 CFR Parts 100-177 Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-11 Standard Specification for Wire Cloth Sieves for Testing Purposes (DOD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Attn. Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

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UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification, Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Purity. The purity of the material shall be not less than 95 percent by weight when tested in accordance with 4.3.4.1.

3.2 Particle size. The material shall pass completely through a 1410-micron (No. 14) sieve when tested in accordance with 4.3.4.2.

3.3 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the precautions of the Material Safety Data Sheet. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.4 Material safety data sheet (MSDS). In accordance with 29 CFR 1910, a Material Safety Data Sheet (MSDS) shall be prepared and submitted in accordance with FED-STD-313 (see 4.2 and 6.4).

3.5 Workmanship. The material shall be uniform in quality and free from foreign materials that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Toxicological formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.3, 6.3, and 6.4).

4.3 **Quality conformance inspection.**

4.3.1 **Lot.** A lot shall consist of the lithium hydride produced by one manufacturer in not more than 24 consecutive hours under essentially the same conditions and with no changes in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall be a lot.

4.3.1.1 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

4.3.2 **Sampling.**

4.3.2.1 **Sampling for examination of product.** A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, inspection level II.

4.3.2.2 **For test.** A sample shall be taken from each lot, the sample size to be calculated as one-tenth of the square root of the number of containers in the lot raised to the next highest whole number. If there are fewer than three containers in a lot, each container shall be sampled. In all other cases, no fewer than three containers shall be selected. From each container in the sample, a representative 500-gram (g) specimen shall be taken and placed in a separate, clean, dry container which is labeled to identify the lot and container represented.

4.3.3 **Examination.**

4.3.3.1 **Examination of product.** Each lot shall be inspected in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 **Sample preparation.** Equal portions of each specimen selected in accordance with 4.3.2.2 shall be thoroughly mixed to form a composite specimen. The composite specimen shall be subjected to the tests specified in 4.3.4.

4.3.3.3 Classification of defects.

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4.3.3.3.1 Preparation for delivery.

Categories	Defects
Critical	
None	
Major	AQL 2.5 percent defective
101	Containers incorrect
102	Containers damaged or leaking
103	Closure of containers incorrect
104	Quantity packaged per container incorrect
105	Marking missing or illegible

4.3.4 Tests. The composite specimen prepared in accordance with 4.3.3.2 shall be subjected to the tests specified herein. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.3.4.1 Purity. Assemble the apparatus as shown on figure 1, attaching the various parts to suitable supports. Fill the gas-measuring buret (D) with hydrogen-saturated water. Adjust the position of the thermometer and the buret delivery tube (L) so that they reach within six millimeters (mm) of the bottom of the reaction flask (N). Turn the three-way stopcock (H) to position 1. Open stopcock (M), raise the leveling bulb (A) until the meniscus of the water in the gas-measuring buret (D) is at the 0-milliliter (ml) mark, and turn the stopcock (H) to position 2. Transfer 0.15 g of the specimen to the reaction flask (N), which should be completely dry. Close stopcock (M) and add about 25 ml of hydrogen saturated water to the buret (L). Read the temperature of the thermometer and turn the three-way stopcock (H) to position 3. Lower the leveling bulb (A) to produce a slightly reduced pressure in the reaction flask (N), open the stopcock (M) and allow 20 ml of water to fall onto the specimen in the reaction flask (N), about 2.0 ml at a time, thereby regulating the rate of the reaction. As the evolution of gas proceeds, lower the leveling bulb (A) so the water level in the bulb is always slightly lower than the level of water in the gas-measuring buret (D). When the reaction is complete, cool the solution in the reaction flask (N) to the temperature of (F_2) prior to the reaction and keep at this temperature for 10 to 15 minutes. Adjust the level of the water in the leveling bulb (A) to the same height as the meniscus of the water in the gas-measuring buret (D) and turn the three-way stopcock (H) to position 2. Record the volume of gas in the gas-measuring buret (D) to the nearest 0.1 ml, read the temperature of the gas from the thermometer (F_1) to the nearest 0.1° C, and the barometric pressure to the nearest 0.1 mm of mercury, making any necessary temperature correction.

Calculate the percent of lithium hydride as follows:

$$\text{Percent lithium hydride (\%)} = 0.1274\ (B - V)\ (P - p)/W\ (273.1 + T)$$

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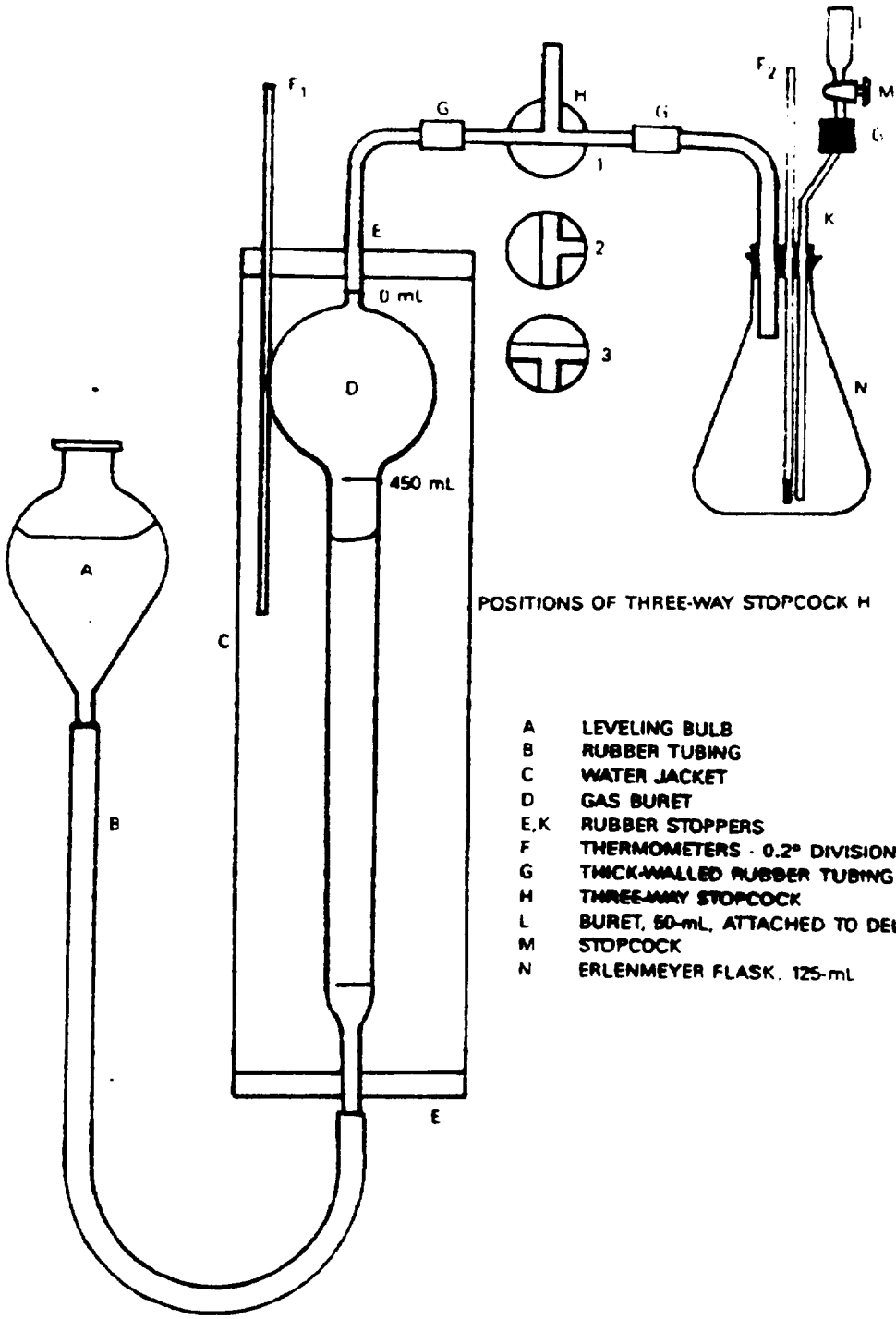


FIGURE 1. Gas evolution apparatus.

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where:

B = volume of gas in the buret at temperature T, in ml

V = volume of water added to the reaction flask, in ml

P = barometric pressure in mm of mercury, corrected to 0 deg C

p = vapor pressure of water at temperature T, in mm of mercury

W = weight of specimen used, in grams

T = temperature reading on thermometer (F_1), in deg C

4.3.4.2 **Particle size.** Place approximately 50 g of the specimen in a 1410-micron (No. 14) sieve conforming to ASTM E-11 assembled with a bottom pan and top cover. Shake for 10 minutes by hand or five minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Observe if any of the material is retained on the sieve.

4.4 Acceptance or rejection. If the composite specimen selected as specified in 4.3.3.2 meets all the requirements of sections 3 and 5, the lot represented shall be accepted. Otherwise, the lot shall be rejected.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Same as for level C.

5.1.2 **Level C.** Lithium hydride shall be packaged in accordance with 49 CFR 173.206 in metal cans in quantities up to but not exceeding 3.8 liters (1 gallon) capacity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Same as for level C.

5.2.2 **Level C.** Unless otherwise specified (see 6.2), lithium hydride packaged as specified in 5.1.2 and in bulk quantities shall be packed in shipping containers in such a manner as to ensure safe transportation by common carrier. Containers used shall comply with National Motor Freight Classification Rules, Uniform Freight Classification Ratings, Rules, and Regulations, or regulations of other carriers applicable to the mode of transportation. Containers used for all shipments shall comply with 49 CFR 173.206.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and 49 CFR 173.206.

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5.3.1 **Precautionary marking.** All individual containers shall be marked with the following precautionary markings:

"DANGER! FLAMMABLE

REACTS VIOLENTLY WITH WATER

LIBERATING AND IGNITING HYDROGEN

MAY CAUSE BURNS

KEEP FROM ANY POSSIBLE CONTACT WITH WATER.

KEEP CONTAINER TIGHTLY CLOSED AND AWAY FROM HEAT.

DO NOT GET IN EYES OR ON SKIN.

WEAR GOGGLES AND DRY GLOVES WHEN HANDLING.

IN CASE OF FIRE, SMOTHER WITH DRY SODA ASH - NEVER WATER OR CHEMICAL FIRE EXTINGUISHERS.

IN CASE OF ACCIDENT, GET FIRST AID AND CALL A PHYSICIAN IMMEDIATELY."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The material covered by this specification is intended for use in generating hydrogen.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Whether the contractor is responsible for inspection (see 4.1).

d. Whether the contractor should use his own laboratory or other suitable facility (see 4.1).

e. Any special packaging and packing requirements (see 5.1).

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f. Any special shipping containers (see 5.2).

g. Any special marking requirements (see 5.3).

6.3 **Caution.** Lithium hydride is a dangerous fire hazard. It is easily ignited by friction, static electricity, or by the heat generated upon reaction with small amounts of water. The hydrogen evolved upon decomposition or reaction is highly flammable and explosive. The caustic residue of lithium hydroxide is dangerous to personnel. Extreme care must be taken in sampling, testing, use and disposal. Operator shall use gloves, protective clothing, face mask, and conduct all operations in a hood or behind a protective shield.

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Subject term (key word) listing.

Analysis
Hazards
Particle size
Purity

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - OS
(Project No. 6810-N005)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-12039D(OS)	2. DOCUMENT DATE (YYMMDD) 14 NOVEMBER 1990

3. DOCUMENT TITLE
LITHIUM HYDRIDE, TECHNICAL (METRIC)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
TOD RICKS	301-743-4358/4510	364-4358/4510
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL ORDNANCE STATION (3730C)	# YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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